SE Lab 1

Requirements Engineering & UML Use-Case Modelling
Campus Placement & Internship Pipeline

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1. Requirements Table

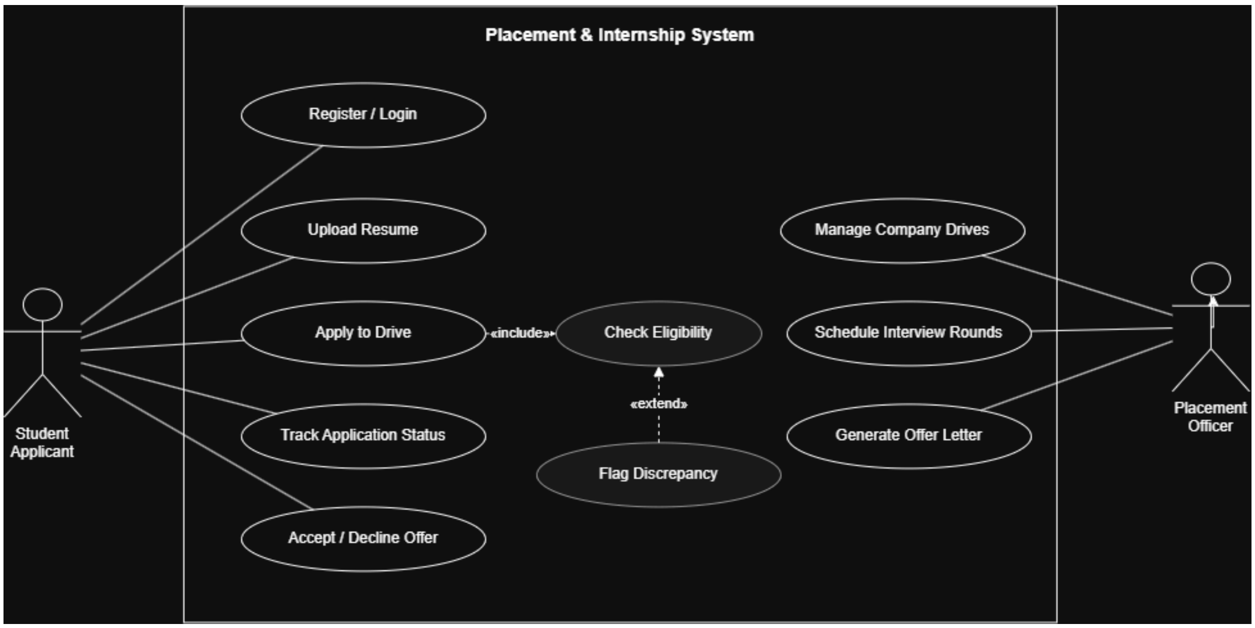
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes (PDF format), extract skill tags, and match profiles against active company eligibility rules.	High	**Pass:** Ineligible applicants are filtered out with clear discrepancy reasons. **Fail:** Applicant with active backlog bypasses eligibility filter.	Manual resume screening across hundreds of drives is the biggest bottleneck for the placement cell; automated parsing is the core value proposition.
FR-002	Functional	The system shall apply multi-tier CGPA and active-backlog filters per drive, since each company sets its own eligibility thresholds.	High	**Pass:** A student below a drive's CGPA cutoff is excluded from that drive's applicant pool but remains visible for other drives they qualify for. **Fail:** Cutoff applied globally instead of per-drive.	Eligibility isn't uniform — a student can qualify for one company and not another; filtering must be drive-scoped, not student-scoped.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-003	Functional	The system shall manage multi-round interview queues, allowing the Placement Officer to advance, reject, or reschedule candidates at each round independently.	High	**Pass:** A candidate rejected in Round 1 is automatically removed from the Round 2 queue. **Fail:** Rejected candidate still appears in later rounds.	Drives commonly run 3–5 rounds (OA, technical, HR); queue state must be consistent across rounds to avoid manual reconciliation errors.
FR-004	Functional	The system shall generate and issue offer letters to shortlisted candidates and record accept/decline responses.	Medium	**Pass:** Offer status updates to "Accepted" only after student confirms; unconfirmed offers auto-expire after a set window. **Fail:** Offer marked accepted without student action.	Prevents disputes over offer timelines and gives the placement cell an auditable record of who accepted what.
FR-005	Functional	The system shall notify students of real-time application status changes (applied, shortlisted, rejected, offer issued) via email/dashboard.	Medium	**Pass:** Status change triggers a notification within 5 minutes. **Fail:** Student status changes silently with no notification.	Reduces the placement cell's support-ticket load from students asking "what's my status."

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Non-Functional (Security)	The system shall encrypt all student academic records and offer documents at rest using AES-256.	High	**Pass:** Benchmarking tests confirm target latency and security standards under simulated peak load.	Academic records and offer letters are sensitive PII/financial data; a breach has legal and reputational consequences for the university.
NFR-002	Non-Functional (Performance)	The system shall support at least 500 concurrent student sessions during peak placement season (e.g., result-announcement windows) with page load latency under 2 seconds.	High	**Pass:** Load testing at 500 concurrent users shows p95 latency < 2s. **Fail:** Response times degrade beyond 2s or requests time out under load.	Placement season causes traffic spikes when results/interview calls drop simultaneously; the system must not buckle exactly when it matters most.

2. UML Use-Case Diagram

Actors: Student Applicant (primary), Placement Officer (secondary). Relationships shown: «include» from Apply to Drive to Check Eligibility, and «extend» from Flag Discrepancy to Check Eligibility.



3. Actors & Use Cases

Actors

- **Student Applicant (Primary):** Handles profile registration, resume submission, and drive applications; monitors progress and finalizes offer decisions.
- **Placement Officer (Secondary):** Oversees recruitment events, coordinates interview logistics, and facilitates the issuance of employment documentation.

Use Cases

- Authentication (Register/Login)
- Resume Data Submission
- Drive Participation («include» Check Eligibility)
- Qualification Verification (Check Eligibility)
- Discrepancy Reporting («extend» Check Eligibility)
- Application Status Monitoring
- Offer Decision Management
- Recruitment Drive Administration

- Interview Sequence Scheduling
- Offer Documentation Generation

4. Use-Case Flow Specification: Apply to Drive

Use Case: Apply to Drive
Primary Actor: Student Applicant
Secondary Actor: Placement Officer (Process Administrator)
Preconditions

- User has authenticated via a valid session.
- Parsed skill profile and PDF resume are successfully indexed.
- A recruitment window is currently open for the selected company.

Postconditions

- Formal application instance is persisted in the database.
- Verification logs for eligibility are recorded.
- Stakeholder notification dispatch is completed.

Main Success Scenario

- Candidate navigates the catalog of available recruitment drives.
- Candidate initiates the application protocol for a specific entry.
- System invokes *Check Eligibility* to validate CGPA and backlog status against drive-specific rules.
- The automated validation results in a passing state.
- Entry status is updated to "Applied" within the system.
- Automated confirmation is transmitted to the applicant.
- The profile is populated in the administrator's review queue.

Alternate Flow — A1: Eligibility Failure
3a. The system identifies a discrepancy regarding academic cutoffs or backlog constraints.
3b. *Flag Discrepancy* is triggered to document the specific non-compliance rationale.
3c. Record is finalized with an "Ineligible" classification.
3d. System provides the discrepancy detail to the user and redirects to the dashboard.
3e. Scenario terminates without data transmission to the placement queue.